

Propylene Glycol — USP/Food Grade

PRODUCT DATA SHEET
FT-PG-USP-2024

1. PRODUCT IDENTIFICATION

CAS No.	57-55-6	EC No.	200-338-0
Formula	$\text{C}_3\text{H}_8\text{O}_2$ — $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{OH}$		Molar mass 76.09 g/mol
Chemical family	Aliphatic diol	Grade	USP / Food grade E1520

Available pack sizes: 25 L can (HDPE) | 200 L drum (HDPE) | 1000 L IBC (HDPE)

2. PHYSICO-CHEMICAL PROPERTIES

Property	Value
Physical state	Colourless viscous liquid
Odour	Practically odourless, slightly sweet
Boiling point	188.2 °C
Flash point	99 °C (closed cup)
Freezing point	-59 °C
Density at 20°C	1.036 g/mL
Viscosity at 20°C	56 mPa·s
Solubility in water	Miscible in all proportions
Hygroscopicity	Hygroscopic
Refractive index nD20	1.4326

3. ANALYTICAL SPECIFICATIONS

Parameter	Specification
Purity (GC)	≥ 99.5%
Water	≤ 0.20%
Density at 20°C	1.035–1.037 g/mL
Refractive index	1.4320–1.4330
Acidity (as lactic acid)	≤ 0.007%
Arsenic (As)	≤ 1.5 ppm
Lead (Pb)	≤ 5 ppm
Chlorides (Cl ⁻)	≤ 10 ppm
Ethylene glycol	≤ 0.06%
Diethylene glycol	≤ 0.10%

4. APPLICATIONS AND USES

- Pharmaceutical solvent (excipient in syrups and injectables)
- Humectant in cosmetics and personal care
- Food additive (humectant E1520 — bread, cakes, colourings)
- Non-toxic heat transfer fluid (food-industry chillers)
- Carrier for flavours and fragrances in e-liquids
- Non-corrosive antifreeze in food-grade HVAC systems

5. HAZARD INFORMATION

This product is not classified as hazardous under the CLP Regulation (EC) No 1272/2008.

6. PRECAUTIONARY STATEMENTS

Code	Precautionary statement
P260	Avoid breathing vapours at high temperature
P273	Avoid release to the environment

7. STORAGE AND SHELF LIFE

Store under ambient conditions (5–30°C), away from direct light and moisture. Keep in tightly closed containers. Compatible with most materials. Shelf life: 24 months. Check regularly for the absence of microbial contamination.

8. TRANSPORT AND REGULATIONS

Not subject to ADR regulations in packages of 1000 L or less

9. DETAILED APPLICATION NOTES

Material compatibility

Before use at scale, check compatibility with seals, pipework and containers. Give preference to materials tested and validated by ChemCorp Industries. Refer to the chemical compatibility tables available on our technical portal (portail.chemcorp.fr/compat). If in doubt, contact our Technical Application service on +33 4 72 XX XX XX.

Quality control on receipt

On receipt, check the integrity of the packaging and that the batch numbers match the accompanying certificate of analysis. Carry out a density and pH check where available. In the event of an anomaly, keep the original packaging and contact our quality department within 48 hours. Procedure ref.: QC-RECEP-003.

Waste management

Waste from this product must be managed in accordance with local hazardous waste regulations (Directive 2008/98/EC). Contact an approved contractor for disposal. European Waste Code (EWC): 06 01 04* for acid solutions, 07 01 01* for halogenated solvents, 07 01 04* for other organic solvents. Never discharge into drains or watercourses.

First aid (operational summary)

INHALATION: Move the person to fresh air, in a semi-seated position. If breathing is difficult, administer O₂ and call the emergency services (15 in France). SKIN CONTACT: Remove contaminated clothing. Rinse with plenty of water for at least 15 minutes. EYE CONTACT: Rinse with water for at least 15 minutes, holding the eyelids open. Consult an ophthalmologist. INGESTION: Do not induce vomiting. Rinse the mouth. Call the Poison Control Centre (European number: 0800 59 59 59).

Environmental data (in-house results)

BOD₅ (5 days): tested per ISO 5815. COD: tested per ISO 15705. Log Kow (n-octanol/water partition coefficient): measured per OECD 117. Aerobic biodegradability: tested per OECD 301B. Aquatic toxicity (Daphnia magna): EC₅₀ per OECD 202. For the complete numerical values, refer to section 12 of the Safety Data Sheet.

Additional regulatory information

This product is registered under REACH: registration number available on request. It is / is not on the SVHC candidate list (last ECHA update: Jan 2025). IED Directive: applicable to installations using this product in quantities above 200 t/year. French ICPE: check the classification under the ICPE nomenclature (heading 1175, 4110 or 4XXX depending on the product). Certifications: USP/NF, BP/EP, FCC, E1520 | Halal & Kosher on request | ISO 22000 | FSSC 22000.